Crabs on camera: technical report

Gina Brøten

2026-08-05

Background

The basis of this project is rooted in the work done in the masters thesis by Brøten (2026). The aim of the project was using the underutuilized resource of underwater video from surveys and coastal monitoring programs from the Insitute of Marine Research

Where the general idea was that coastal monitoring programs and mulitple surveys at the Institue of Marine Research applied video surveys to monitor coastal habitats.

Brown crabs

The data collected and used to train this model

Prerequisites

Need to have Python installed. `raw_to_yolo.ipynb` will check for missing packages and install them automatically (see "Chunk 0").

Ideally a server if you want to train your own model.

To keep the repo lightweight, `.ipynb` outputs are stripped on commit via `nbstripout`. One-time setup after cloning:

```
pip install nbstripout
nbstripout --install
```

Data

The repo only ships a small example/test dataset (`data/annotations/test_annotations`) so the scripts can be run and verified without downloading everything.

The full training dataset (~8,000 images) used in the original masters project is hosted externally: [**LINK TO ADD**]. Download it and point `CONFIG["ANNOTATIONS_FILE"] / CONFIG["IMAGES_DIR"]` in `raw_to_yolo.ipynb` at the local copy to reproduce the full training run.

From raw annotations to YOLO-ready data

To

Brøten G. Crabs on camera: Improving the monitoring of the brown crab (*Cancer pagurus*) with machine learning and Baited Remote Underwater Video (BRUV). Master's thesis, University of Bergen, 2026. https://hdl.handle.net/11250/5537346.